

FDS EXPERIMENT- 1

Exp No:1.a Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

```
[ ] ▶ import pandas as pd
import matplotlib.pyplot as plt

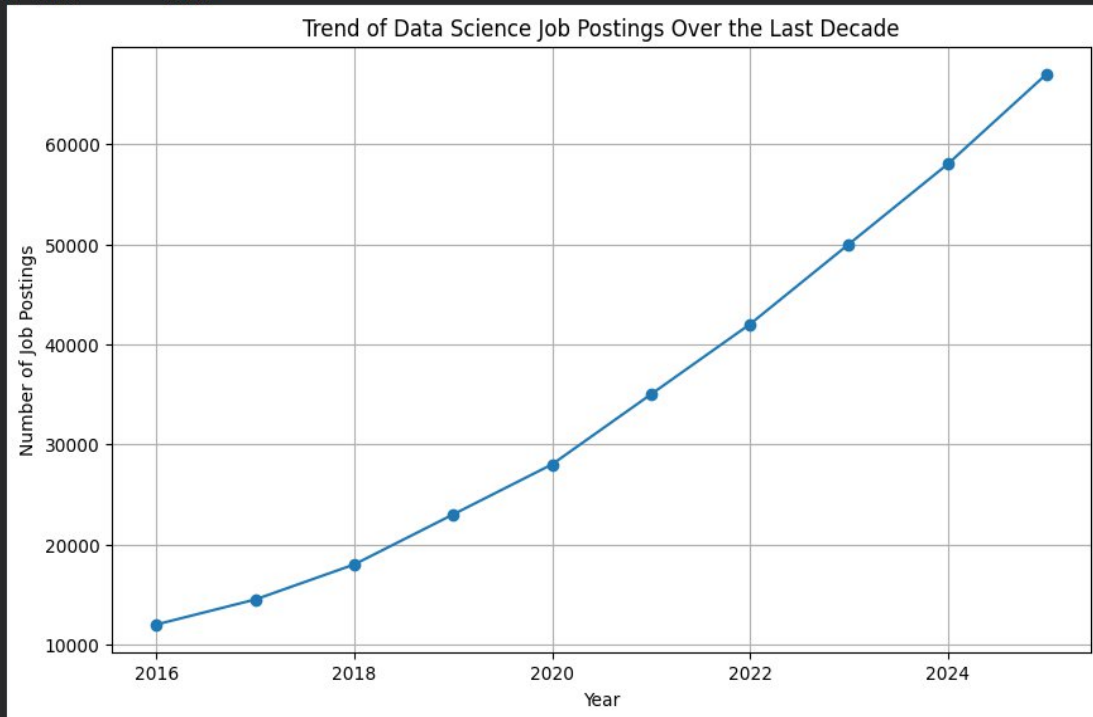
years = list(range(2016, 2026))
job_postings = [12000, 14500, 18000, 23000, 28000, 35000, 42000, 50000, 58000, 67000]

df = pd.DataFrame({
    'Year': years,
    'Job_Postings': job_postings
})

print(df)

plt.figure(figsize=(10, 6))
plt.plot(df['Year'], df['Job_Postings'], marker='o')
plt.title('Trend of Data Science Job Postings Over the Last Decade')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.grid(True)
plt.show()
```

```
... 0 2016 12000
1 2017 14500
2 2018 18000
3 2019 23000
4 2020 28000
5 2021 35000
6 2022 42000
7 2023 50000
8 2024 58000
9 2025 67000
```



Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

```

import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Role': [
        'Data Analyst', 'Data Scientist', 'Data Engineer',
        'Data Analyst', 'Data Scientist', 'Data Engineer',
        'Data Analyst', 'Machine Learning Engineer',
        'Data Scientist', 'Data Engineer', 'Data Analyst',
        'Machine Learning Engineer', 'Data Scientist', 'Data Analyst',
        'Data Engineer'
    ]
}

df = pd.DataFrame(data)

role_counts = df['Role'].value_counts()

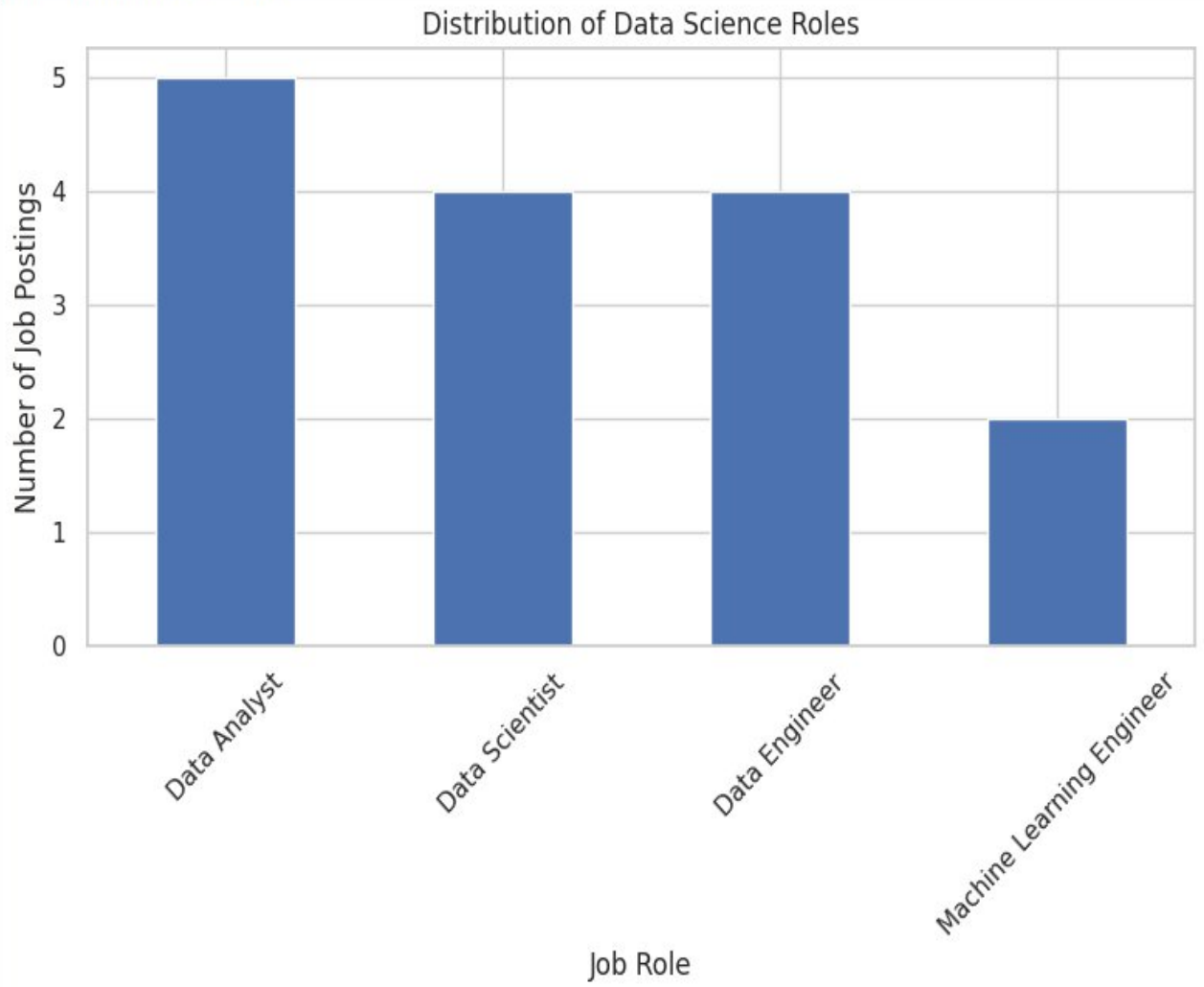
print(role_counts)

plt.figure(figsize=(8, 6))
role_counts.plot(kind='bar')
plt.title('Distribution of Data Science Roles')
plt.xlabel('Job Role')
plt.ylabel('Number of Job Postings')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

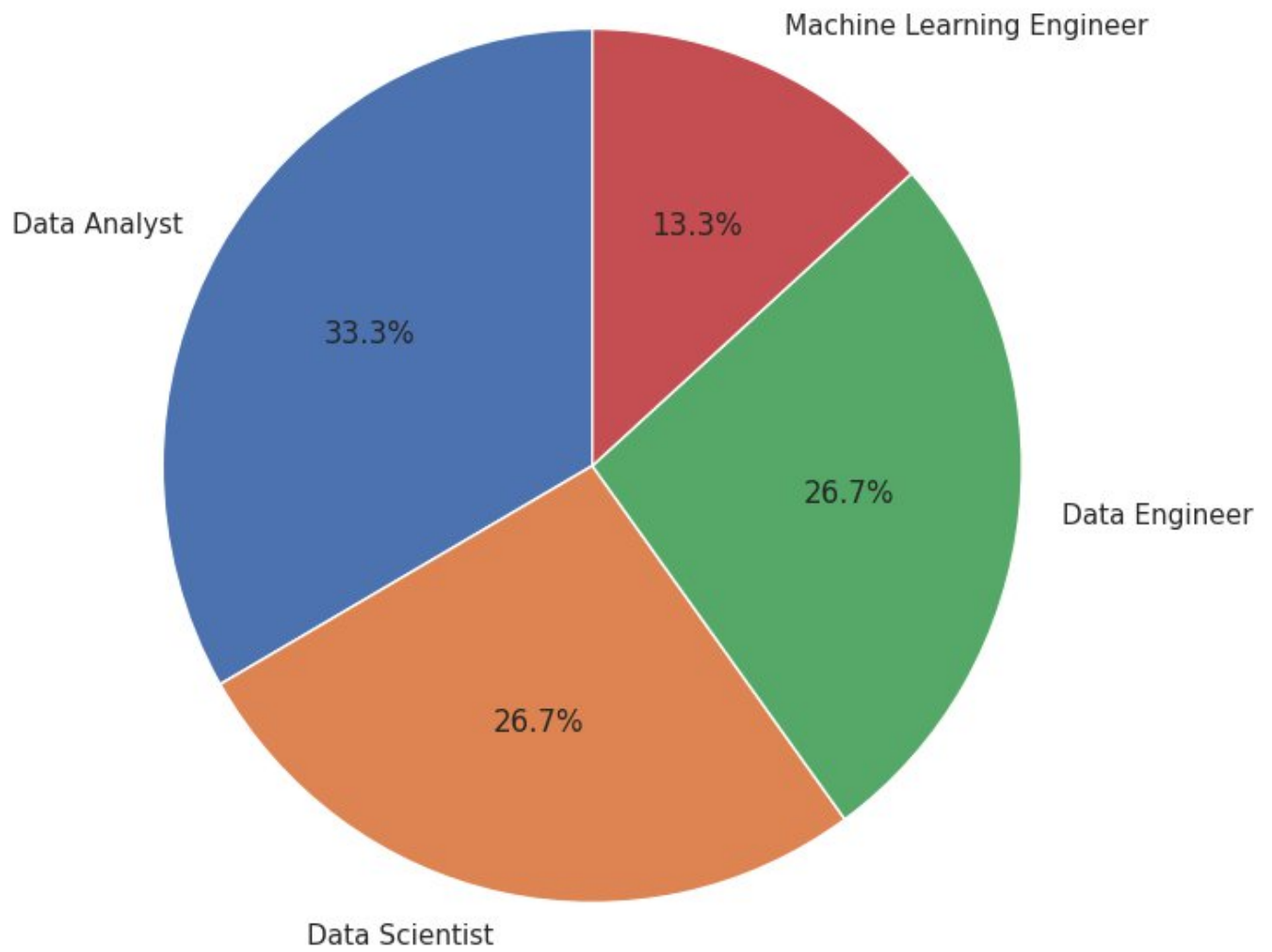
plt.figure(figsize=(8, 8))
role_counts.plot(kind='pie', autopct='%1.1f%%', startangle=90)
plt.title('Distribution of Data Science Roles')
plt.ylabel('')
plt.show()

```

```
... Role
Data Analyst      5
Data Scientist    4
Data Engineer     4
Machine Learning Engineer  2
Name: count, dtype: int64
```



Distribution of Data Science Roles



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

```
import pandas as pd
import json

structured_data = pd.DataFrame({
    'ID': [101, 102, 103],
    'Name': ['Arun', 'Bala', 'Kumar'],
    'Age': [20, 21, 19]
})

print("STRUCTURED DATA")
print(structured_data)
print("\nCharacteristics: Rows and columns, fixed structure, easy to analyze.")

semi_structured_data = [
    {"ID": 101, "Name": "Arun", "Department": "CSE"},
    {"ID": 102, "Name": "Bala", "Department": "ECE"}
]

print("\nSEMI-STRUCTURED DATA")
print(json.dumps(semi_structured_data, indent=2))
print("\nCharacteristics: Uses keys/tags, flexible structure, example: JSON.")

unstructured_data = [
    "Arun is a CSE student.",
    "Bala likes Python."
]

print("\nUNSTRUCTURED DATA")
print(unstructured_data)
print("\nCharacteristics: No fixed structure, example: text, images, audio and video.")
```

*** STRUCTURED DATA

	ID	Name	Age
0	101	Arun	20
1	102	Bala	21
2	103	Kumar	19

Characteristics: Rows and columns, fixed structure, easy to analyze.

SEMI-STRUCTURED DATA

```
[
  {
    "ID": 101,
    "Name": "Arun",
    "Department": "CSE"
  },
  {
    "ID": 102,
    "Name": "Bala",
    "Department": "ECE"
  }
]
```

Characteristics: Uses keys/tags, flexible structure, example: JSON.

UNSTRUCTURED DATA

```
['Arun is a CSE student.', 'Bala likes Python.']
```

Characteristics: No fixed structure, example: text, images, audio and video.

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.



```
!pip install cryptography
```

```
from cryptography.fernet import Fernet
```

```
key = Fernet.generate_key()
```

```
f = Fernet(key)
```

```
token = f.encrypt(b"Rajalakshmi Engineering College")
```

```
print("Original Data:")
```

```
print(b"Rajalakshmi Engineering College")
```

```
print("\nEncryption Key:")
```

```
print(key)
```

```
print("\nEncrypted Data:")
```

```
print(token)
```

```
decrypted_data = f.decrypt(token)
```

```
print("\nDecrypted Data:")
```

```
print(decrypted_data)
```

```
... Requirement already satisfied: cryptography in /usr/local/lib/python3.12/dist-packages (49.0.0)
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from cryptography) (2.1.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=2.0.0->cryptography) (3.0)
Original Data:
b'Rajalakshmi Engineering College'

Encryption Key:
b'ceSIhD0FgJzut5HtqUGY1-qmDCjBygYxbCp6UaRcSUI='

Encrypted Data:
b'gAAAAABqgGWWfo1-4RzXs8Qm00y7R54jjBFEBa3fGxaYTbsH204367_BBzmKS9gpd0epJHicP4vIaWQjBGZp9QyKSLYhgKbKH_FwRkx9uQqD6MX1VECvG54='

Decrypted Data:
b'Rajalakshmi Engineering College'
```